

Principal Component Analysis — by hand

Four points, two variables, one pencil. No computer needed.

1 Data & means

	x	y
A	0	1
B	1	2
C	3	3
D	4	2
Σ	8	8

$$\bar{x} = 8 / 4 = 2$$
$$\bar{y} = 8 / 4 = 2$$

2 Centre the data & build the covariance matrix

obs	x	y	$x - \bar{x}$	$y - \bar{y}$	products		
			a	b	a · a	b · b	a · b
A	0	1	-2	-1	4	1	2
B	1	2	-1	0	1	0	0
C	3	3	1	1	1	1	1
D	4	2	2	0	4	0	0
Σ	8	8	0	0	10	2	3

✓ **check:** the centred columns must sum to 0.

$$\text{Var}(x) = \Sigma a^2 / (n-1) = 10/3$$

$$\text{Var}(y) = \Sigma b^2 / (n-1) = 2/3$$

$$\text{Cov}(x,y) = \Sigma ab / (n-1) = 3/3 = 1$$

$$C = \begin{pmatrix} 10/3 & 1 \\ 1 & 2/3 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 10 & 3 \\ 3 & 2 \end{pmatrix} = \frac{1}{3} S$$

Shortcut: $S = X^T X$ for the centred data matrix X . Dividing by $n-1$ (not n) is “Bessel’s correction”; it only rescales the eigenvalues, never the eigenvectors. Work with S , divide by 3 at the end.

A little history. Karl Pearson (1901) posed it geometrically as the “line of closest fit” to points in space; Harold Hotelling (1933) named the “principal components” and computed them with an iterative power method on a desk calculator. Both did precisely what this poster does — sums of squares and products, then a characteristic equation — by hand.

3 Eigenvalues

$$\det(S - \lambda I) = 0$$

$$(10-\lambda)(2-\lambda) - 3 \cdot 3 = 0$$

$$\lambda^2 - 12\lambda + 11 = 0$$

$$(\lambda - 11)(\lambda - 1) = 0$$

$$\lambda_1 = 11 \quad \lambda_2 = 1$$

✓ trace $10+2 = 12 = 11+1$ · det $20-9 = 11 = 11 \cdot 1$
(of C : $11/3$ and $1/3$)

4 Eigenvectors

$$\lambda = 11: (S - 11I)v = 0$$

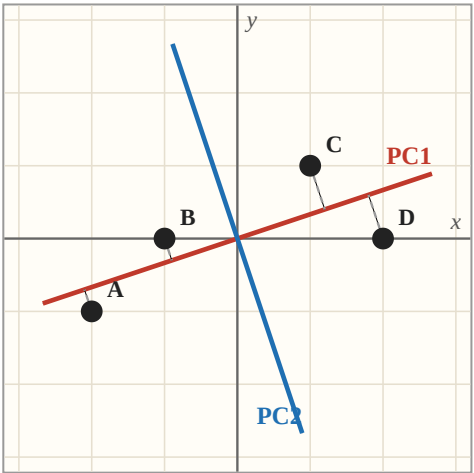
$$-v_1 + 3v_2 = 0 \rightarrow v = (3, 1)$$

$$\text{length } \sqrt{10} \rightarrow \text{PC1} = (3, 1)/\sqrt{10}$$

$$\lambda = 1: 9v_1 + 3v_2 = 0 \rightarrow v = (1, -3)$$

$$\text{PC2} = (1, -3)/\sqrt{10}$$

✓ perpendicular: $3 \cdot 1 + 1 \cdot (-3) = 0$



Centred data with the two principal axes. Dashed: projection onto PC1.

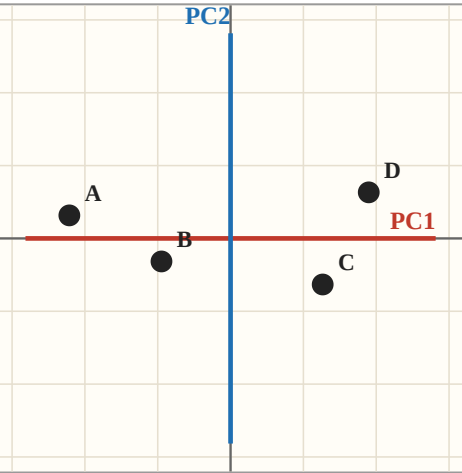
5 Scores (project the data)

$$\text{PC1} = (3a + b)/\sqrt{10}$$

$$\text{PC2} = (a - 3b)/\sqrt{10}$$

obs	PC1	PC2
A	-2.21	0.32
B	-0.95	-0.32
C	1.26	-0.63
D	1.90	0.63
Σ score ²	11	1

✓ $\Sigma \text{score}^2 = \text{eigenvalue}$: $(49+9+16+36)/10 = 11$ and $(1+1+4+4)/10 = 1$



Same points, rotated into the PC1–PC2 frame.

6 Variance explained

$$\text{PC1: } 11 / (11+1) = 91.7 \% \quad \text{PC2: } 1/12 = 8.3 \%$$



One axis keeps almost all the information: 2-D → 1-D.

Recipe. centre → covariance → eigen-decomposition → sort by λ → project.
Data: A(0,1) B(1,2) C(3,3) D(4,2) — numbers chosen so the eigenvalues (11, 1) are whole.

Extension — three variables & the cubic

Same recipe, one more dimension: the characteristic equation is now a cubic. Data: A(0,3,1) B(1,1,3) C(3,3,4) D(4,1,0), variables x, y, z.

1 Centre & products

$\bar{x} = \bar{y} = \bar{z} = 8/4 = 2$ (a = x-2, b = y-2, c = z-2)

obs	raw			centred			products					
	x	y	z	a	b	c	a·a	b·b	c·c	a·b	a·c	b·c
A	0	3	1	-2	1	-1	4	1	1	-2	2	-1
B	1	1	3	-1	-1	1	1	1	1	1	-1	-1
C	3	3	4	1	1	2	1	1	4	1	2	2
D	4	1	0	2	-1	-2	4	1	4	-2	-4	2
Σ	8	8	8	0	0	0	10	4	10	-2	-1	2

✓ centred columns sum to 0. Three variables → 3 squares + 3 cross-products (the matrix is symmetric).

2 Scatter matrix S (and covariance C = S/3)

$S = \begin{pmatrix} 10 & -2 & -1 \\ -2 & 4 & 2 \\ -1 & 2 & 10 \end{pmatrix}$ C = 1/3S, eigenvalues of C = λ/3

Row i, column j = Σ (variable i)(variable j) from the Σ row above. Diagonal = squares, off-diagonal = cross-products.

3 The characteristic cubic

$\det(S - \lambda I) = 0 \implies \lambda^3 - (\text{tr } S)\lambda^2 + (\Sigma \text{ minors})\lambda - \det S = 0$

$\text{tr } S = 10 + 4 + 10 = 24$

minors: $(10 \cdot 4 - (-2)^2) + (10 \cdot 10 - (-1)^2) + (4 \cdot 10 - 2^2) = 36 + 99 + 36 = 171$

$\det S = 10(4 \cdot 10 - 2 \cdot 2) - (-2)((-2) \cdot 10 - 2 \cdot (-1)) + (-1)((-2) \cdot 2 - 4 \cdot (-1))$
 $= 10 \cdot 36 + 2 \cdot (-18) - 1 \cdot 0 = 324$

$\lambda^3 - 24\lambda^2 + 171\lambda - 324 = 0$

Coefficients are “sum of the roots”, “sum of products of pairs” and “product of the roots” (Vieta) — so they double as checks at the end.

Recipe (any size). centre → S = X^TX → det(S-λI)=0 → solve for λ → (S-λI)v=0 → sort by λ → project.

Numbers chosen so that λ = 12, 9, 3 and the eigenvectors are small whole-number vectors.

4 Solve the cubic by hand

Rational-root test. S is positive semi-definite, so every root is > 0 and < tr S = 24. An integer root must divide 324 = 2²·3⁴: candidates 1, 2, 3, 4, 6, 9, 12, 18.

p(1) = 1-24+171-324 = -176 ✗

p(2) = 8-96+342-324 = -70 ✗

p(3) = 27-216+513-324 = 0 ✓

Deflate with **Horner** (synthetic division by λ-3):

coeffs	1	-24	171	-324
×3		3	-63	324
sum	1	-21	108	0

$\lambda^2 - 21\lambda + 108 = 0 \rightarrow \lambda = (21 \pm \sqrt{(441-432)})/2 = (21 \pm 3)/2$

$\lambda = 12, 9, 3$

✓ sum 12+9+3 = 24 = tr S · products 12·9·3 = 324 = det S · pairs 108+36+27 = 171

5 Eigenvectors

For each λ, S-λI is singular: its rows are all ⊥ to the eigenvector, so the **cross product of two rows** is the eigenvector.

λ = 12: S-12I: row1=(-2,-2,-1), row3=(-1,2,-2)

row₁ × row₃ = (6, -3, -6) ∝ (2, -1, -2), |v| = 3

→ **PC1 = (2, -1, -2)/3**

λ = 9: S-9I: row1=(1,-2,-1), row2=(-2,-5,2)

row₁ × row₂ = (-9, 0, -9) ∝ (1, 0, 1), |v| = √2

→ **PC2 = (1, 0, 1)/√2**

λ = 3: S-3I: row1=(7,-2,-1), row2=(-2,1,2)

row₁ × row₂ = (-3, -12, 3) ∝ (1, 4, -1), |v| = 3√2

→ **PC3 = (1, 4, -1)/(3√2)**

✓ S·PC1: (10·2+2·2, -4·4-4, -2·2-20) = (24, -12, -24) = 12·(2, -1, -2) · dot products between vectors: 2+0-2 = 0, 2-4+2 = 0, 1+0-1 = 0

6 Scores

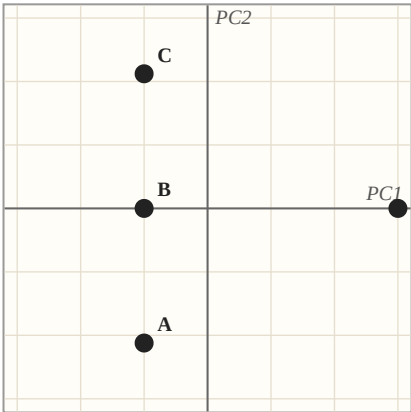
PC1 = (2a - b - 2c)/3

PC2 = (a + c)/√2

PC3 = (a + 4b - c)/(3√2)

obs	PC1	PC2	PC3
A	-1.00	-2.12	0.71
B	-1.00	0.00	-1.41
C	-1.00	2.12	0.71
D	3.00	0.00	0.00
Σ score ²	12	9	3

✓ Σ score² = λ, e.g. PC2: 4.5 + 0 + 4.5 + 0 = 9



First two components: 3-D → 2-D.
PC1 isolates D; PC2 separates A from C.

7 Variance explained

12/24 = 50% · 9/24 = 37.5% · 3/24 = 12.5%

PC1 50 %

PC2 37.5 %

Two components keep 87.5 % of the variance.

If the roots aren't whole numbers? A symmetric 3×3 always has three real roots: use Cardano's trigonometric formula, or Newton's $\lambda \leftarrow \lambda - p(\lambda)/p'(\lambda)$. From degree 5 up no formula exists (Abel–Ruffini), so Hotelling iterated: multiply a vector by S repeatedly — it swings onto PC1 — subtract it, repeat.